Statistical Learning Theory in Deep Learning

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Statistical Learning Theory (STL) aims to predict performance of a machine learning model on unseen data. This notes explains the applications and limitations of STL in deep learning.

Formally, let $S = \{(x_i, y_i)\}_{i=1}^m$ be the dataset where $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$ for all i, and each pair (x_i, y_i) is i.i.d. sampled from an unknown distribution $\mathcal{D}$. Let $f : \mathcal{X} \to \mathcal{Y}$ be a model. On each sample (x_i, y_i), f takes x_i as input and returns a prediction $f(x_i)$. The mismatch between the prediction $f(x_i)$ and the target y_i is measured by the loss function $l(f(x_i), y_i) \in \mathbb{R}$. l is smaller if $f(x_i)$ is closer to y_i. We define the empirical risk as $R^{\text{emp}}(f) = \sum_{i=1}^m l(y_i, f(x_i))$ and the true risk as $R^{\text{true}}(f) = \mathbb{E}_{(x_i,y_i)\sim\mathcal{D}}[l(y_i, f(x_i)]$.

Empirical risk measures how well the model fits the observed data, while true risk measures how well the model fits the unseen data. A model with smaller true risk implies that it will achieve more accurate prediction on unseen data. However, the true risk cannot be computed directly since it takes expectation over the unknown data distribution $\mathcal{D}$.

For any given f, $R^{\text{emp}}(f)$ is an unbiased estimator of $R^{\text{true}}(f)$. Therefore, we can construct an upper bound for the gap $R^{\text{true}}(f) - R^{\text{emp}}(f)$ to estimate $R^{\text{true}}(f)$. Such bound is called the *generalization bound*.

Empirical Risk Minimization and Uniform Convergence

Since true risk is not optimizable, we typically obtain the model $\hat{f}$ through Empirical Risk Minimization (ERM), i.e., solving the following problem

$$\hat{f} = \text{argmin}_{f\in\mathcal{F}} R^{\text{emp}}(f)$$

where $\mathcal{F}$ is the *hypothesis space*, the set of all models that we search from.

However, the $\hat{f}$ obtained from ERM may not minimize true risk, i.e., its good prediction on observed data may not be generalized on unseen data. The question is, under what condition can we obtain an $\hat{f}$ that also optimizes true risk? The uniform convergence theory developed by Vapnik and Chervonenkis [1] answered this question.

Consider the binary classification problem where the range of the labels is $\mathcal{Y} = \{-1, 1\}$. Define the binary loss $l(f(x), y) = \mathbf{1}[f(x) \neq y] \in \{0, 1\}$. Then for any $\epsilon > 0$, the following bound holds [1]

$$P\left(\sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right| > \epsilon\right) < 4S_{\mathcal{F}}(2m)e^{-\epsilon^2 m/8} \tag{1}$$

Or equivalently, for any $0 < \delta < 1$,

$$P\left(\sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right| < \sqrt{\frac{8}{m}\log\frac{4S_{\mathcal{F}}(2m)}{\delta}}\right) > 1 - \delta \tag{2}$$

Here $S_{\mathcal{F}}$ is the *growth function* of $\mathcal{F}$, which is a measure of the size of $\mathcal{F}$. $\mathcal{F}$. Formally,

$$S_{\mathcal{F}}(m) = \max_{(x_1,x_2,...,x_m)\in\mathcal{X}^m} |\{(f(x_1), f(x_2), ..., f(x_m)) : f \in \mathcal{F}\}|$$

Note that each $f(x_i) \in \{-1, 1\}$, $\{(f(x_1), f(x_2), ..., f(x_m)) : f \in \mathcal{F}\}$ is the set of classification outputs that all models in $\mathcal{F}$ can possibly generate. $(x_1, x_2, ..., x_m)$ is chosen to maximize the cardinality of this set.

Clearly we have $S_{\mathcal{F}}(m) \leq 2^m$. If $S_{\mathcal{F}}(m) = 2^m$, we say $\mathcal{F}$ *shatters* m points. The Vapnik-Chervonenkis (VC) dimension of $\mathcal{F}$, denoted as h, is defined as the maximum number of points it can shatter, i.e., $h = \max\{m : S_{\mathcal{F}}(m) = 2^m\}$. By Sauer's Lemma [9] we have $S_{\mathcal{F}}(2m) \leq \left(\frac{2em}{h}\right)^h$. Therefore, (2) implies

$$P\left(\sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right| < \sqrt{\frac{8}{m}\log\frac{4h\log\frac{2em}{h}}{\delta}}\right) > 1 - \delta \tag{3}$$

The bound (3) shows that $\sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right| \to 0$ in probability as $m \to \infty$. Note that this convergence holds uniformly for all f in $\mathcal{F}$, including the empirical risk minimizer $\hat{f}$. Therefore, with large enough m, $R^{\text{true}}(f)$ will be close to $R^{\text{emp}}(f)$, and $\hat{f}$ will also nearly optimize true risk.

Vacuousness

One issue of uniform convergence in deep learning is *vacuousness*. Since the loss is binary, trivially we have $\sup_{\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right| \leq 1$. Thus, the bound (3) will be *vacuous* if $\sqrt{\frac{8}{m}\log\frac{4h\log\frac{2em}{h}}{\delta}} \geq 1$, i.e., its prediction for the gap $\sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right|$ is not meaningful. If we choose a hypothesis space $\mathcal{F}$ with large h, then m should be equally large to make (3) non-vacuous.

For deep neural network models, h grows with a complexity $O(WL\log(W))$, where W is the number of parameters and L is the number of layers [10]. This means that h can be extremely large and can easily make (3) vacuous. However, empirical results have shown that these models achieves not only near zero training error but also small testing error, i.e., the gap $|R^{\text{true}}(f) - R^{\text{emp}}(f)|$ is actually small. This means that the uniform convergence bound (5) may be too vacuous to explain the good generalization performance of deep neural networks.

One reason for this mismatch between theory and practice is because of the regularization techniques like regularizer, dropout, weight decay in stochastic gradient descent used in training deep neural networks [3]. These regularization techniques allow us to search for models in a subset $\mathcal{F}'$ of the set $\mathcal{F}$. The size of $\mathcal{F}'$ can be much smaller than the size of $\mathcal{F}$, potentially resulting in a tighter bound. However, how to derive a tighter enough generalization bound that explains the performance of deep neural networks is still underexplored in theory.

Structural Risk Minimization

Structural Risk Minimization (SRM) serves as the theoretical foundation for regularization techniques. It explains why adjusting the size of $\mathcal{F}$ helps optimizing the true risk.

It easy to see that for any $f \in \mathcal{F}$,

$$R^{\text{true}}(f) \leq R^{\text{emp}}(f) + \sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right| \tag{4}$$

where $\sup_{f\in\mathcal{F}} \left|R^{\text{true}}(f) - R^{\text{emp}}(f)\right|$ can be estimated by the confidence interval of (3).

Suppose we choose $\mathcal{F}$ from a set of nested hypothesis spaces $\mathcal{F}_1 \subset \mathcal{F}_2 \subset ... \subset \mathcal{F}_n$. Let h_i be the VC dimension of $\mathcal{F}_i$, it is easy to show that $h_1 \leq h_2 \leq ... \leq h_n$. A hypothesis space $\mathcal{F}$ with VC dimension h means that by choosing models in $\mathcal{F}$, we can correctly classify at most h samples $\{(x_i, y_i)\}_{i=1}^h$ with arbitrary labels $\{y_i\}_{i=1}^h \in \{-1, +1\}^h$. Hence, for some datasets with m samples, we have $\min_{f\in\mathcal{F}_i} R^{\text{emp}}(f) \leq 1 - \frac{h_i}{m}$ when $h_i < m$ and $\min_{f\in\mathcal{F}_i} R^{\text{emp}}(f) = 0$ when $h_i \geq m$. This implies the minimal empirical risk tends to decrease as h increases.

Moreover, (3) shows that the confidence interval $\sqrt{\frac{8}{m}\log\frac{4h\log\frac{2em}{h}}{\delta}}$ increases with h (suppose m and δ are fixed). Consequently, the relationship between the bound on true risk (4) and the VC dimension of $\mathcal{F}$ can be illustrated as Figure 1.

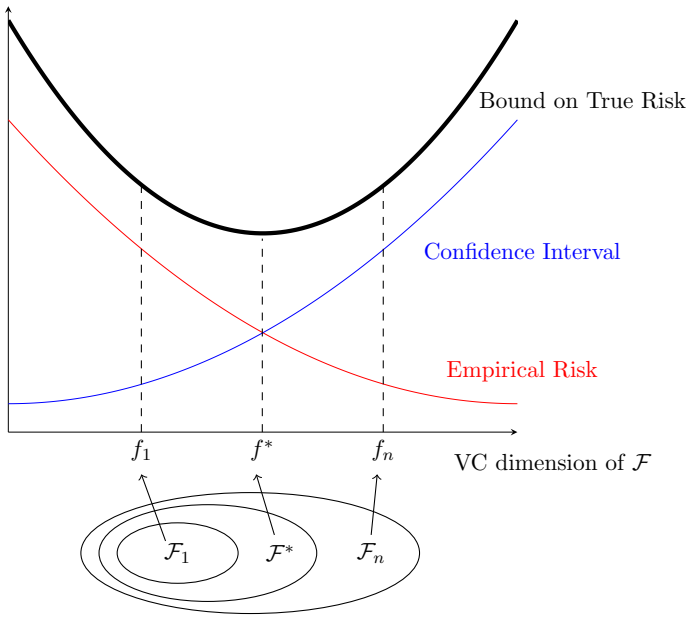

Figure 1: Structural Risk Minimization

Let f^* be the minimizer of the bound on true risk. Our goal is to search for f^*. The principle of SRM is that, we construct a set of nested hypothesis spaces $\mathcal{F}_1 \subset \mathcal{F}_2 \subset ... \subset \mathcal{F}_n$, then choose an $\mathcal{F}_i$ with appropriate size and search for f^* in $\mathcal{F}_i$. $\mathcal{F}_i$ being too small may fail to include f^*, while it being too large may cause f^* difficult to search for – both cases make the bound on true risk large.

VC dimension is one metric to measure the complexity of $\mathcal{F}$, i.e., its ability of correctly classifying the data samples. Other metrics include Rademacher complexity and covering number.

Uniform Generalization Bounds

A uniform generalization bound is a bound that hold for all models in some model space. In PAC bounds, it is typically in the form $P\left(\sup_{f\in\mathcal{F}}\{R^{\text{true}}(f) - R^{\text{emp}}(f)\} < g(\delta)\right) > 1-\delta$ for some hypothesis space $\mathcal{F}$. In PAC-Bayes bounds, when f is considered a stochastic model following a distribution ρ, it is typically of the form $P\left(\forall\rho\in\Theta, \mathbb{E}_{f\sim\rho}[R^{\text{true}}(f) - R^{\text{emp}}(f)] < g(\delta)\right)$ $> 1-\delta$ for some distribution space Θ.

These bounds have weak forms. For PAC bounds, $\forall f\in\mathcal{F}, P\left(R^{\text{true}}(f) - R^{\text{emp}}(f) < g(\delta)\right)$ $> 1-\delta$. For PAC-bayes bounds, $\forall\rho\in\Theta, P\left(\mathbb{E}_{f\sim\rho}[R^{\text{true}}(f) - R^{\text{emp}}(f)] < g(\delta)\right) > 1-\delta$. The difference is that, in the strong form, the probability P is measured over a union of events, each event is about whether a single model holds for the bound. This implies that the bound simultaneously hold for all models. In the weak form, P is measured over only a single event, implying that it holds only for a single model.

A uniform bound is essential in performing SRM. The underlying assumption of Figure 1 is that the bound is true on any models in $\mathcal{F}$. If the bound holds only for a single model, such as f_1, then it may be false on f^* and f_n (in other words, the domain of the curve in Figure 1 is $\{f_1\}$ rather than the entire $\mathcal{F}$). Consequently, such bound is inapplicable to a broader class of models.

Regularization

Regularization is the technique that we implement SRM in practice. There are two types of regularization: explicit and implicit [3].

A typical explicit regularization method is to add one or more regularizer terms to the empirical risk. Formally, let Θ_f be the vector of all parameters of f, and $\|\Theta_f\|_2$ be the L2 norm of Θ_f. We choose a set of positive integers $\{c_i\}_{i=1}^n$ such that $c_1 < c_2 ... < c_n$, and define $\mathcal{F}_i = \{f : \|\Theta_f\|_2^2 \leq c_i\}$. In this way, we construct a set of nested hypothesis spaces $\mathcal{F}_1 \subset \mathcal{F}_2 \subset ... \subset \mathcal{F}_n$. Optimizing the empirical risk by searching for f in $\mathcal{F}_i$ is then formed as the constrained optimization problem

$$\hat{f} = \text{argmin}_{f \in \mathcal{F}} \, R^{\text{emp}}(f) \quad s.t. \, \|\Theta_f\|_2^2 \leq c_i \tag{5}$$

The constraint in (5) can be converted to a Lagrange multiplier, i.e., given c_i, there exists some λ_i such that the following optimization problem

$$\hat{f} = \text{argmin}_{f \in \mathcal{F}} \, R^{\text{emp}}(f) + \lambda_i \|\Theta_f\|_2^2 \tag{6}$$

is equivalent to (5). Also, the sequence $\{\lambda_i\}_{i=1}^n$ satisfies $\lambda_1 > \lambda_2 ... > \lambda_n$. The $\lambda_i \|\Theta_f\|_2^2$ term in (6) is called the *regularizer*. In practice, we implement SRM by trying different λ_i to solve (6), evaluating each obtained $\hat{f}$ on test set, and choosing the one that achieves the highest evaluation performance as the optimal model.

As for implicit regularization, one popular technique is *early stopping* [3]. Suppose we use gradient descents to train deep neural networks. With the number of epochs increases, the training error will decrease, while the testing error may decrease first, then increases, as illustrated in Figure 2 (a). This implies that the optimal model f^* may lie in the middle of searching path. Moderate number of epochs helps approaching f^*, while too many epochs push our searching far away from f^*, as shown in Figure 2 (b) [4]. Early stopping means we stop searching at some thresholds of number of epochs, when observing the testing error starts to increase.

Zhang et al [3] showed that early stopping may not always work. They empirically showed that the testing accuracy of the Alexnet and MLP models on CIFAR 10 constantly increase with number of epochs, providing no threshold for early stopping.

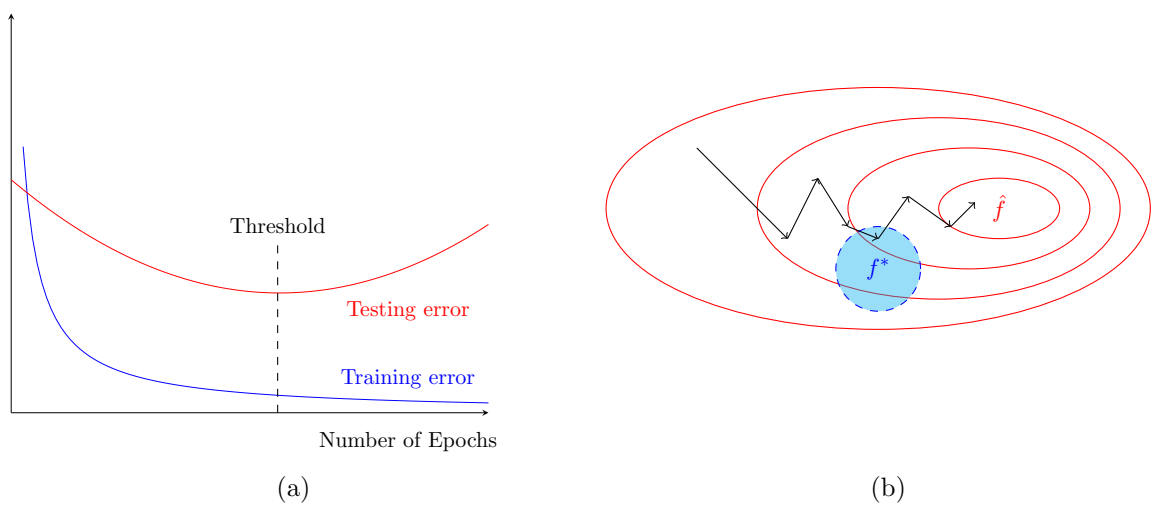

Figure 2: Early Stopping

Bias-Variance Trade-off

In statistics, the Mean Squared Loss (MSE) can be decomposed as bias and variance terms. Given an model f, define the squared loss as $(f(x)-y)^2$, then $R^{\text{true}}(f) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[(f(x)-y)^2]$. It is well known that the optimal model that minimizes $R^{\text{true}}(f)$ is $f^*(x) = \mathbb{E}_{y|x}[y]$. Therefore, for any model f, the excess risk can be expressed as

$$\begin{aligned}
R^{\text{true}}(f) - R^{\text{true}}(f^*) &= \mathbb{E}_{(x,y)}[(f(x)-y)^2] - \mathbb{E}_{(x,y)}[(f^*(x)-y)^2] \\
&= \mathbb{E}_{(x,y)}[f(x)^2 - f^*(x)^2 - 2y(f(x)-f^*(x))] \\
&= \mathbb{E}_x[f(x)^2 - f^*(x)^2 - 2\mathbb{E}_{y|x}[y](f(x)-f^*(x))] \\
&= \mathbb{E}_x[(f(x)-f^*(x))^2]
\end{aligned}$$

Now, suppose f^* is fixed, we consider f a random variable used to estimate f^*. Then we have the following bias-variance decomposition [6, 8].

$$\begin{aligned}
\mathbb{E}_{x,f}[(f(x)-f^*(x))^2] &= \mathbb{E}_x\mathbb{E}_{f|x}[(f(x)-f^*(x))^2] \\
&= \mathbb{E}_x\left[\text{Var}_{f|x}[f(x)-f^*(x)] + \mathbb{E}_{f|x}[f(x)-f^*(x)]^2\right] \\
&= \mathbb{E}_x\text{Var}_{f|x}[f(x)] + \mathbb{E}_x\left(\mathbb{E}_{f|x}f(x) - f^*(x)\right)^2 \\
&= \underbrace{\mathbb{E}_{x,f}\left[(f(x)-\mathbb{E}_{f|x}f(x))^2\right]}_{\text{Variance}} + \mathbb{E}_x\left(\underbrace{\mathbb{E}_{f|x}f(x) - f^*(x)}_{\text{Bias}}\right)^2
\end{aligned}$$

The last equality above uses the fact that $\text{Var}_{f|x}[f(x)] = \mathbb{E}_{f|x}\left[(f(x)-\mathbb{E}_{f|x}f(x))^2\right]$.

Figure 1 shows the bias-variance tradeoff in SRM: Let f be a random model in $\mathcal{F}$. Low complexity of $\mathcal{F}$ represents high bias and low variance, while high complexity of $\mathcal{F}$ represent low bias and high variance. The bias relates to the empirical risk: Low bias means for a random model f, its expectation $\mathbb{E}f$ will be close to f^*, thus any realization of f, when evaluated on datasets generated from $\mathcal{D}^m$, will likely to give a low empirical risk. The variance relates to the size of $\mathcal{F}$: A larger $\mathcal{F}$ may have higher complexity, which enlarges the confidence interval; larger $\mathcal{F}$ also means the searching range for f is larger, implying higher variance.

Thus, the bias-variance trade-off gives a statistical explanation for the U shape curve in Figure 1, in the context of squared loss.

Generalization of Over-parameterized Neural Networks

Zhang et al [3] claimed that, while regularization helps improve generalization performance, it is neither sufficient nor necessary to explain why the generalization gap for deep neural networks is small in practice. Recent study explains this phenomenon in the perspective of interpolation.

Experimental results have show that over-parameterized neural network can easily achieves zero training error [3]. This implies that the model interpolates the dataset perfectly, and achieves the global minimum of empirical risk. As shown in Figure 3 (a), classic generalization theory demonstrates that the testing error will increase when the complexity of hypothesis space $\mathcal{F}$ is beyond some spot. However, experiments show that after reaching the interpolation threshold, the testing error decreases again and gradually converges, as shown in the "double descending U curves" in Figure 3 (b). Here the interpolation threshold refers to as the spot that the training error reaches zero.

Theoretical Tools for the Over-parameterized Models

Classic PAC bounds are typically vacuous for over-parameterized models. In fact, such bounds have been theoretically shown that they cannot be non-vacuous without being dependent on datasets and training algorithms [11]. Suppose we have a training algorithm $\mathcal{A}$ that

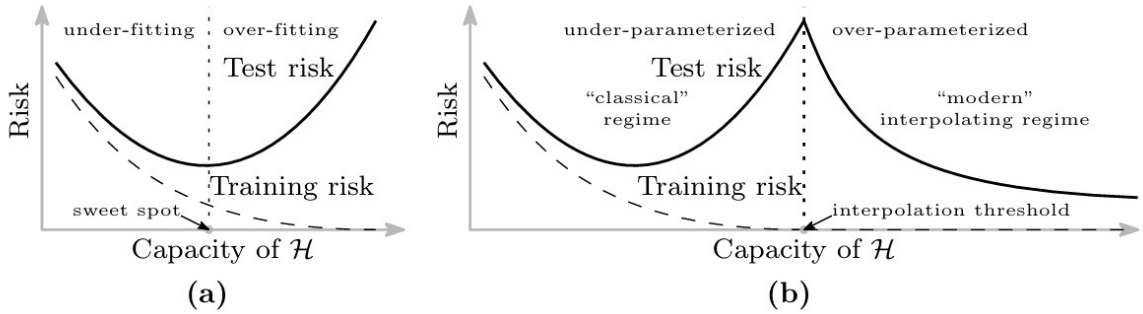

Figure 3: Curves for training risk (dashed line) and test risk (solid line). Figure referred from [7], where $\mathcal{H}$ corresponds to $\mathcal{F}$ in this note.

takes in a dataset S and returns a model in $\mathcal{F}$. The obtained model is said to be dependent on $\mathcal{A}$ and S, which typically occupies a small subset of $\mathcal{F}$ rather than the entire $\mathcal{F}$. The complexity of these subsets is much smaller than the entire $\mathcal{F}$ so that they can result in a tighter bound.

One example of data- and algorithm- dependent bounds is the PAC-Bayes bound by Dziugaite and Roy [12]. They assume the stochastic model follow some distributions, with the prior π and the posterior ρ. They first assume π and ρ to be Gaussian to facilitate computation, then under given π they involve dataset S and training algorithm $\mathcal{A}$ to compute for ρ, and empirically show that such bound is non-vacuous. The issue is that, since ρ is obtained from optimization, in most cases it may not have a closed-form, which does not allow a clean analysis for the bound [13]. An exception is shown by Guo et al. [14] that when the model is linear and under the squared loss, ρ has a closed-form. But this result is not applicable to non-linear deep neural networks.

Regardless of the generalization bounds, other tools for analyzing over-parameterized models include neural tangent kernel (NTK) [15] and mean field analysis [16]. In particular, NTK is adapted to over-parameterized models by assuming that the width of a neural network, rather than the depth, goes to infinity. Under this condition, the kernel of the neural network admits a closed-form solution after training. However, in practice, neural networks are typically of finite width, and the infinite-width assumption of NTK is too extreme to align with practical cases.

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